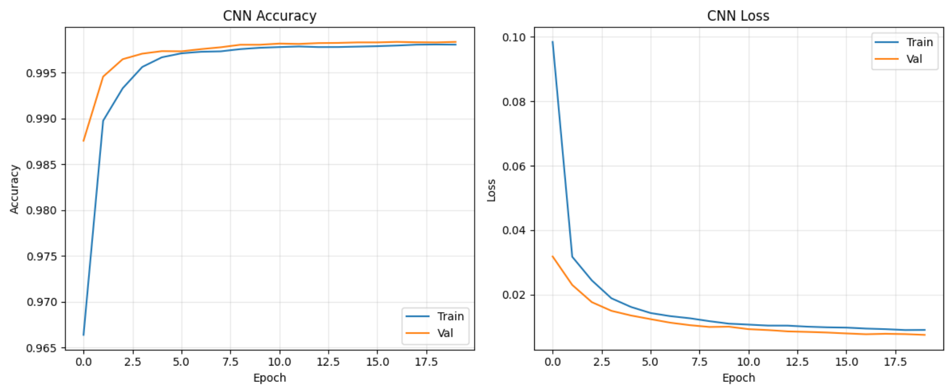
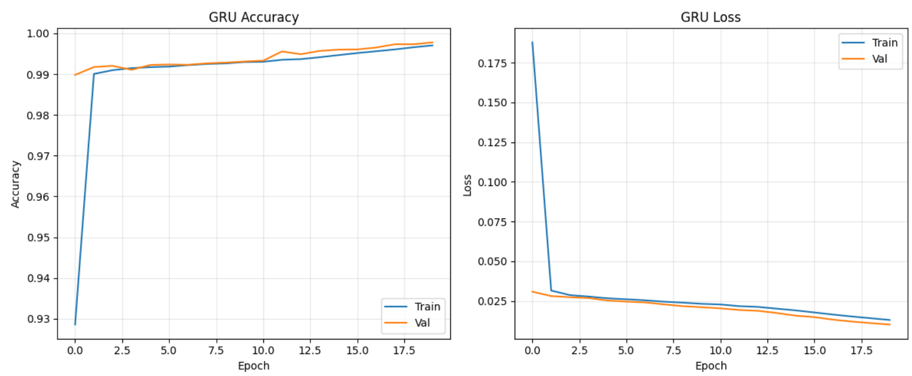
Supplementary Material for High-Throughput and Low-Latency DrDoS Detection Using 2 Quantized ONNX Models

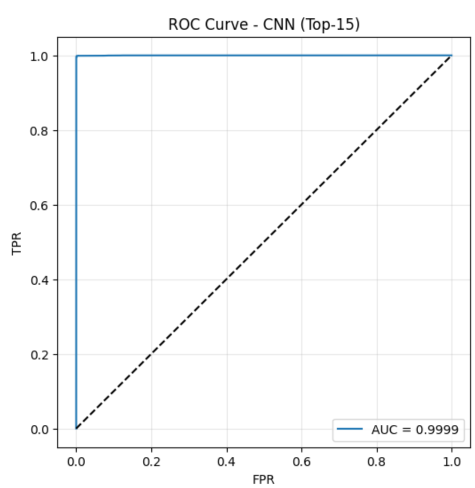
Validation-Training Loss and Accuracy Graphs for the CNN Model



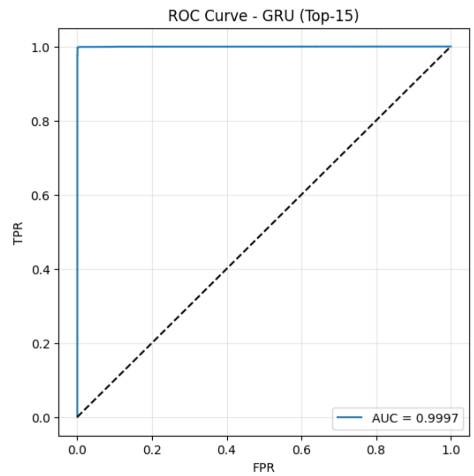
Validation-Training Loss and Accuracy Graphs for the GRU Model



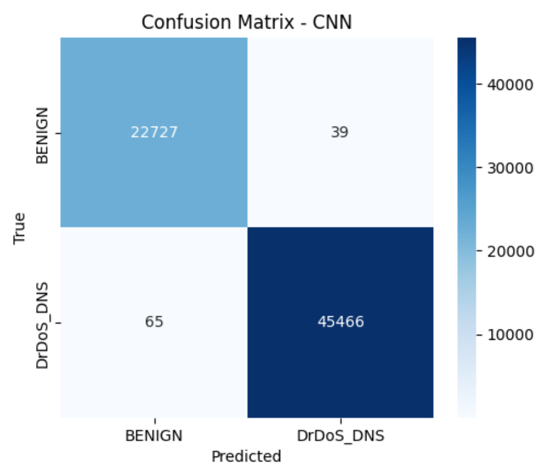
Test ROC Curve for the CNN Model



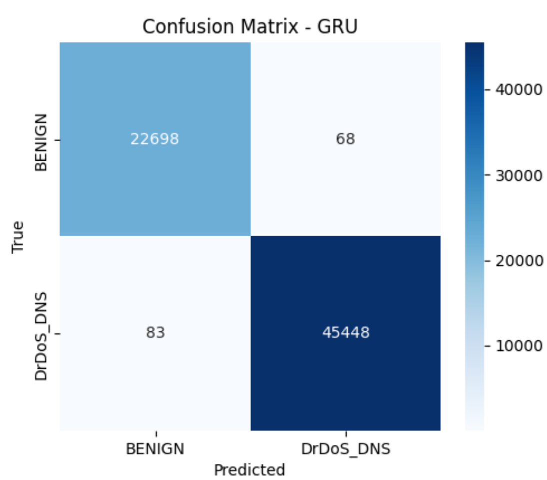
Test ROC Curve for the GRU Model



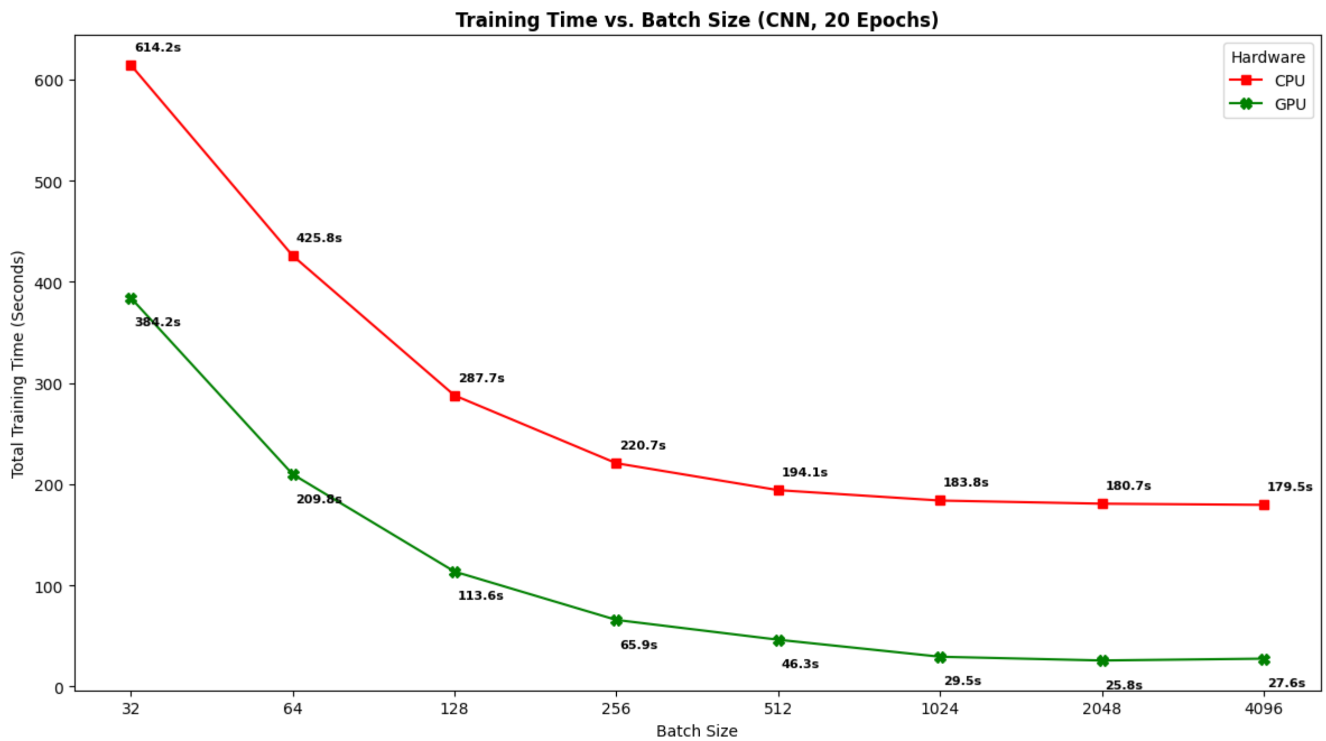
Confusion Matrix for the CNN Model



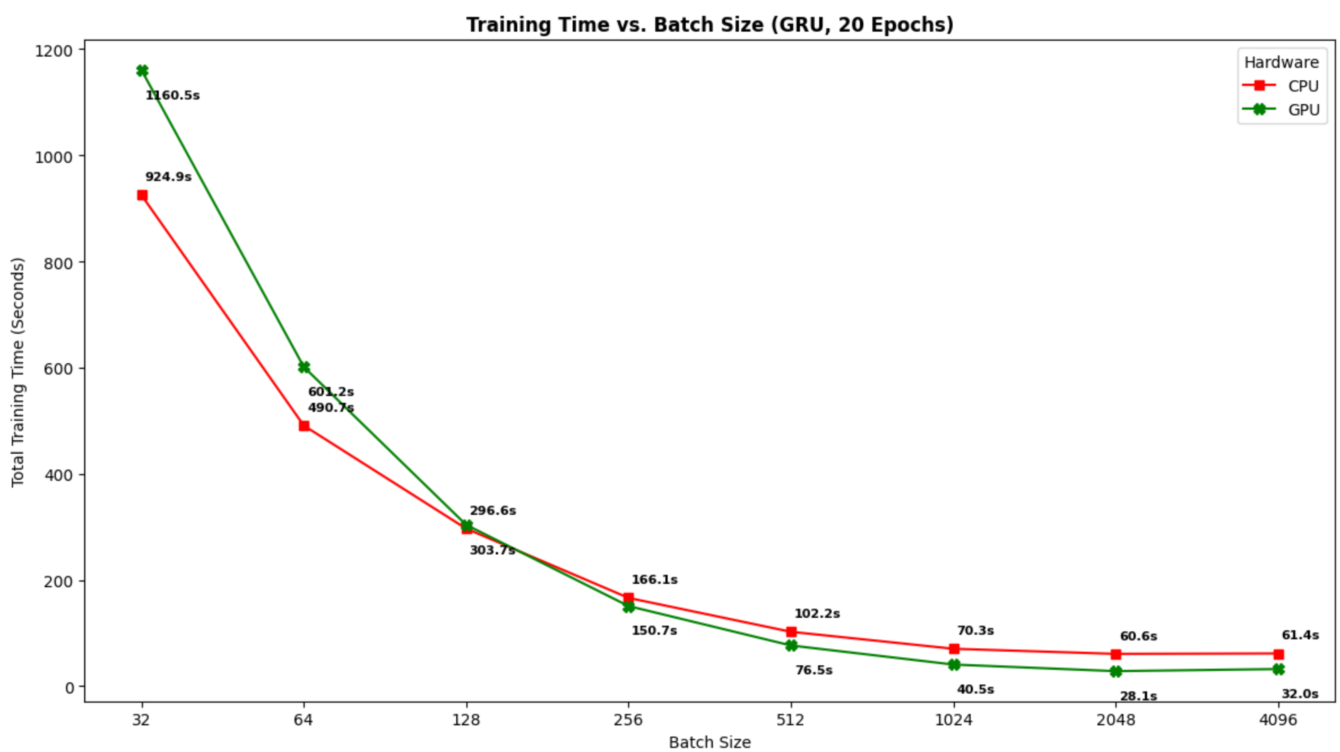
Confusion Matrix for the GRU Model



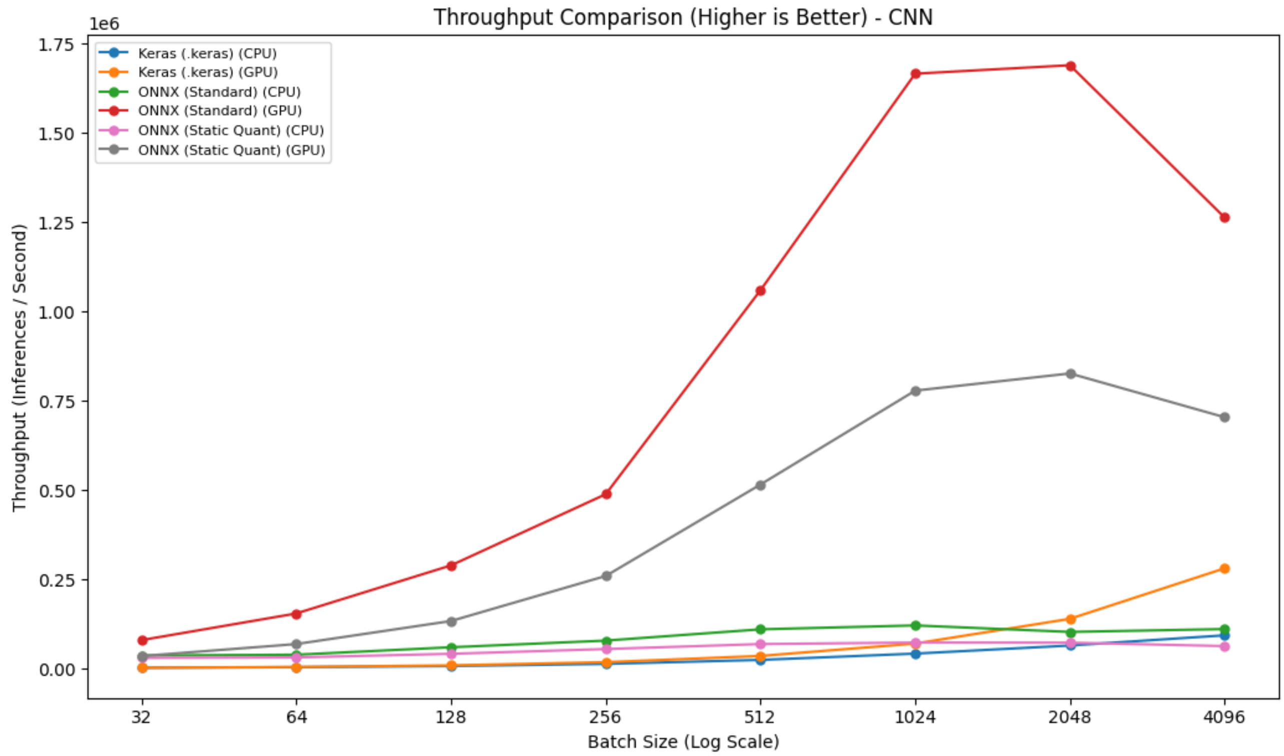
Training Time vs Batch Size (CNN)



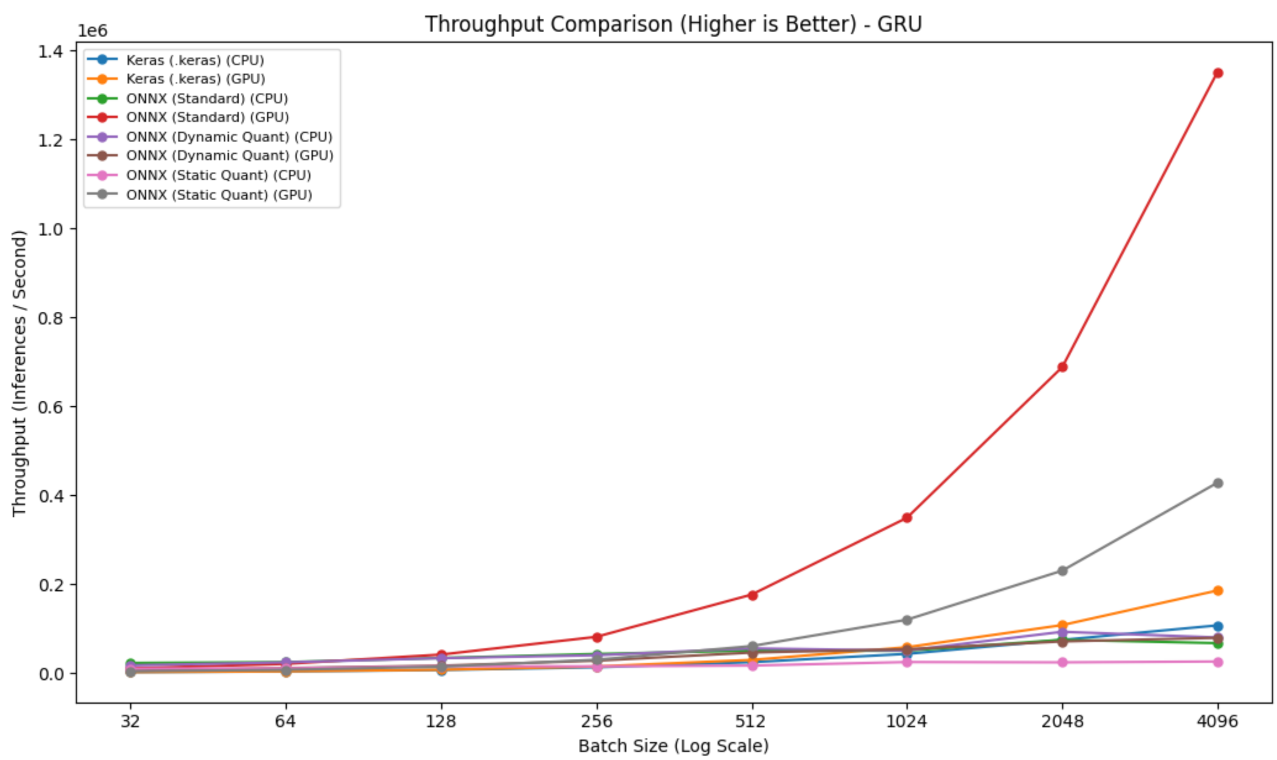
Training Time vs Batch Size (GRU)



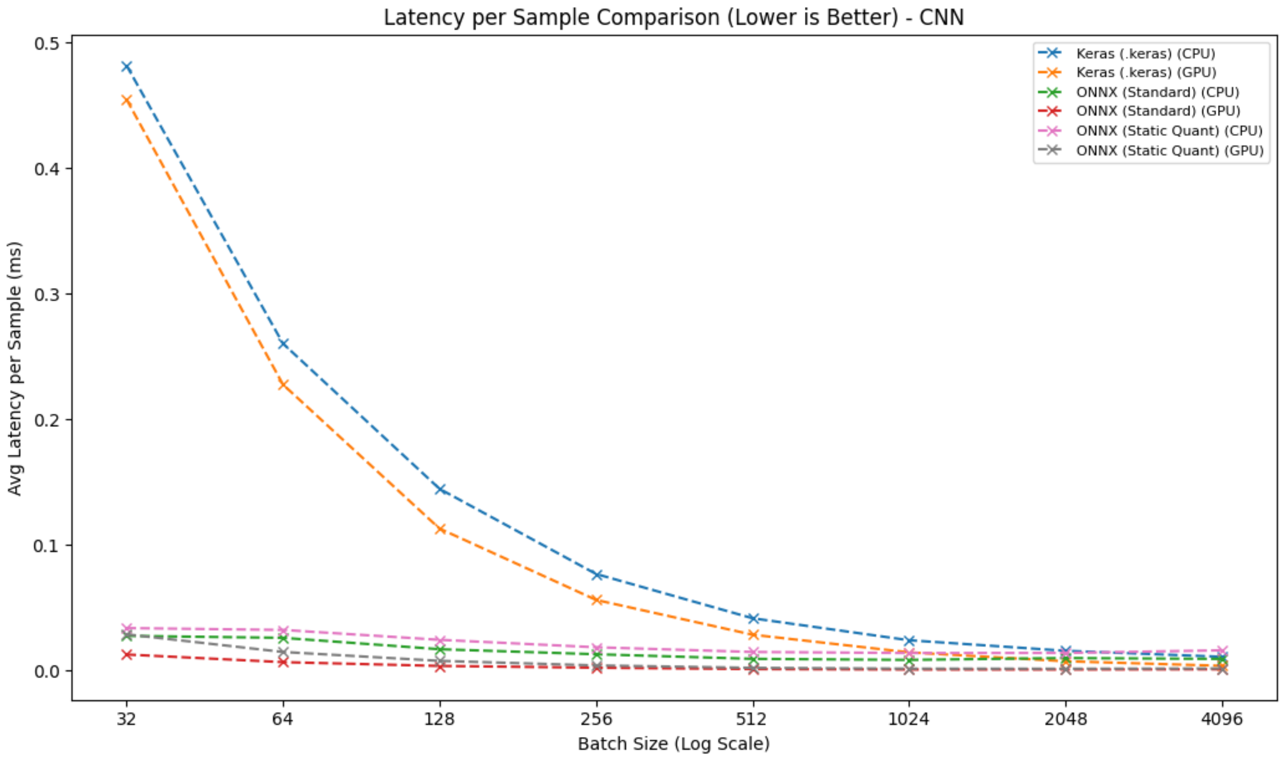
Throughput vs Batch Size (CNN)



Throughput vs Batch Size (GRU)



Latency vs Batch Size (CNN)



Latency vs Batch Size (GRU)

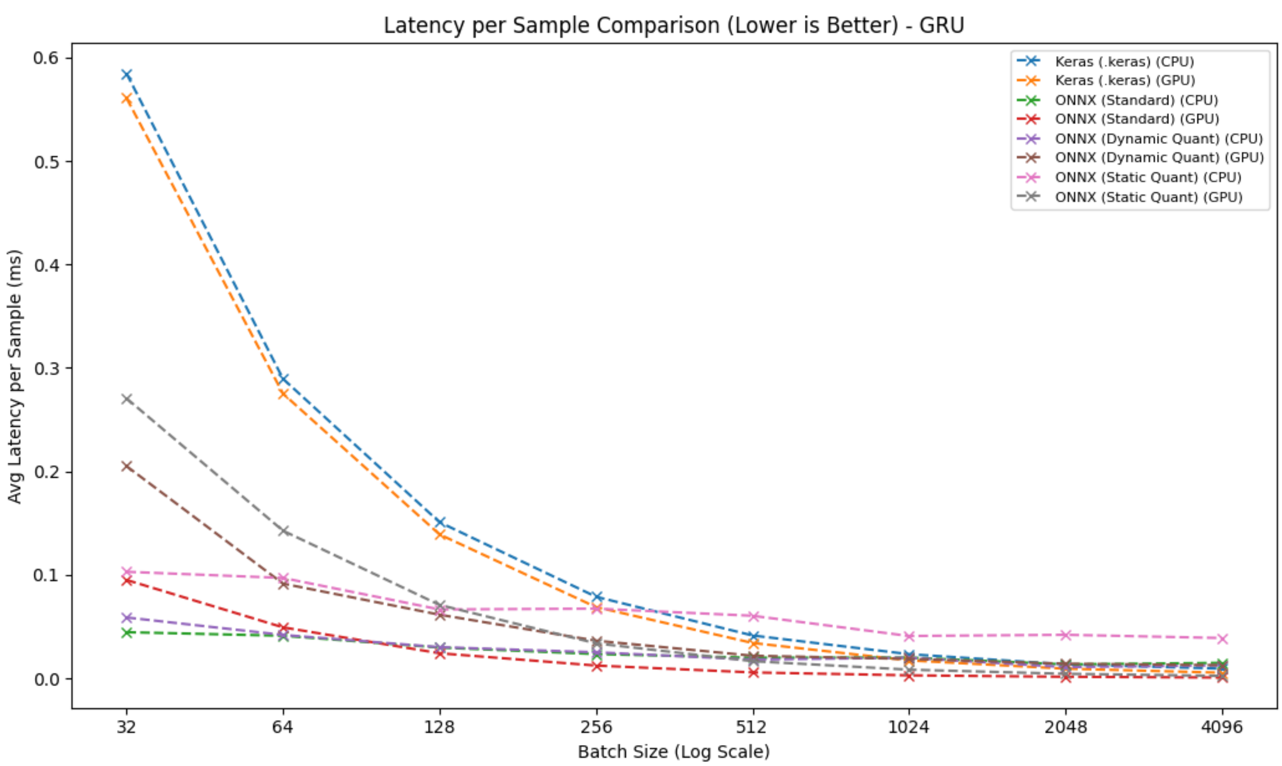


Table 3. CNN Model Comparison Summary.

Model	Environment	Inference Time (µs)	File size (MB)
Keras	CPU	10.7489	0.7086
Keras	GPU	3.5619	0.7086
ONNX	CPU	8.2803	0.226
ONNX	GPU	0.5917	0.226
Static Quantized ONNX	CPU	13.7391	0.1056
Static Quantized ONNX	GPU	1.2097	0.1056

Table 3. GRU Model Comparison Summary.

Model	Environment	Inference Time (µs)	File size (MB)
Keras	CPU	9.3467	0.556
Keras	GPU	5.3923	0.556
ONNX	CPU	13.4195	0.2625
ONNX	GPU	0.7397	0.2625
Dynamic Quantized ONNX	CPU	10.8018	0.2534
Dynamic Quantized ONNX	GPU	12.6615	0.2534
Static Quantized ONNX	CPU	38.9413	0.7065
Static Quantized ONNX	GPU	2.3361	0.7065

LIBRARY VERSIONS (reproducibility)

- python : 3.12.13
- platform : Linux-6.6.122+-x86_64-with-glibc2.35
- numpy : 2.0.2
- pandas : 2.2.2
- scikit_learn : 1.6.1
- scipy : 1.16.3
- matplotlib : 3.10.0
- seaborn : 0.13.2
- tensorflow : 2.20.0
- keras : 3.13.2
- onnx : 1.22.0
- tf2onnx : 1.17.0
- onnxsim : 0.6.5
- protobuf : 5.29.6
- onnxruntime_gpu (GPU inference) : 1.20.1
- onnxruntime_cpu (CPU inference / CNN dynamic-quant): 1.27.0
- cuda : 12.5.1
- cudnn : 9
- gpu : A100 [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
- ONNX Runtime: GPU=onnxruntime-gpu 1.20.1 | CPU=onnxruntime 1.27.0 (iki ayri faz)
- Random seeds: global=42 | split=42 | downsample=27 | variance=1000-1029

All experiments were conducted on Google Colab using an NVIDIA A100 GPU (Linux 6.6.122, x86_64, glibc 2.35; CUDA 12.5.1, cuDNN 9). The software stack was: Python 3.12.13, TensorFlow 2.20.0 (Keras 3.13.2), NumPy 2.0.2, pandas 2.2.2, scikit-learn 1.6.1, SciPy 1.16.3, matplotlib 3.10.0, seaborn 0.13.2, ONNX 1.22.0, tf2onnx 1.17.0, onnxsim 0.6.5, and protobuf 5.29.6. ONNX inference was evaluated with two ONNX Runtime builds in separate sessions: ONNX Runtime-GPU 1.20.1 for GPU benchmarks—executed in a TensorFlow-free session to avoid GPU-memory contention—and ONNX Runtime 1.27.0 (CPU build) for CPU benchmarks, including the CNN dynamic-quantized model whose ConvInteger operator is implemented only in the CPU build. All random seeds were fixed to ensure determinism: a global seed of 42 (NumPy, Python random, and TensorFlow), random_state = 42 for the stratified train/test split, random_state = 27 for downsampling.